

Bistable periodic traveling waves and dynamical stabilization in an integrodifference equation with strong Allee effect and overcompensatory cycles

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Abstract

hello

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1 Introduction

2 Analysis

2.1 Local dynamics

This section contains basic facts about the growth function,

$$g(u; a, r) = u \exp(r(1 - u)(\frac{u}{a} - 1))$$

Desmos plot of graph: <https://www.desmos.com/calculator/bm9fvoluqx>

- Its first derivative is

$$g'(u) = \left(1 + \frac{r(a+1)}{a}u - \frac{2r}{a}u^2\right) \exp(r(1 - u)(\frac{u}{a} - 1))$$

A few special values are:

$$g'(0) = \exp(-r), \quad g'(a) = 1 + r(1 - a), \quad g'(1) = 1 + \frac{r(a-1)}{a}$$

- It has a global maximum:

$$u_m := \operatorname{argmax} g = \frac{1}{4} \left(a + 1 + \sqrt{(a+1)^2 + \frac{8a}{r}} \right)$$

In the limit $u_m \rightarrow (a+1)/2$ as $r \rightarrow \infty$.

- g is increasing on $[0, u_m]$ and decreasing on $[u_m, \infty)$.
- g is monotone on $[0, 1]$ iff. $r \leq \frac{a}{1-a}$.
- (Stability at zero) $u = 0$ is asymptotically stable for all $r > 0$.
- (Stability at threshold) $u = a$ is unstable if $r > 0$ and $a < 1$.
- (Stability at capacity) If $a = 1$ then $g'(1) = 1$ always. So assuming $a < 1$:

$$u = 1 \text{ is stable} \iff |g'(1)| < 1 \iff r < \frac{2}{1-a}$$

$$u = 1 \text{ is unstable} \iff |g'(1)| > 1 \iff r > \frac{2}{1-a}$$

- (Period-doubling cascade) Assuming $a < 1$ then there is a sequence $(r_{a,n})_{n=0}^{\infty}$ such that whenever $r_{a,n} < r < r_{a,n+1}$ then the map $u \mapsto g(u; a, r)$ has a stable 2^n -cycle and unstable 2^k -cycles for all $k < n$. So the first couple values are:

$$r_{a,0} = 0, \quad r_{a,1} = \frac{2}{1-a}, \quad \dots, \quad r_{a,n} \rightarrow r_{a,\infty} < r_{a,\text{collapse}}$$

- (Collapse) If g maps (a, ∞) into $(0, a)$, then $u = 0$ is a global attractor. This occurs when $g(g(u_m)) < a$. So there is a special point r_m depending on a such that when $r > r_m$, there is no longer a positive stable attractor. Thus all the interesting behavior occurs when $0 < r < r_m$.

2.2 Case 1: monotone growth

For this case, we can apply the results of [LO22]

The following is an immediate application of Theorem 3.1:

Proposition 2.1. Suppose $r \leq r_m$. Then there exists a unique c such that there exists a nonincreasing traveling wave $w \in C(\mathbb{R}, [0, 1])$ such that

$$Qw(x) = w(x - c)$$

with $w(-\infty) = 1$ and $w(\infty) = 0$.

2.3 Case 2: nonmonotone growth, stable carrying capacity

2.4 Case 3: nonmonotone growth, stable 2-cycles

When $r_1 < r < r_m^1$ we can apply the results of [BLL18]

2.5 Else: nonmonotone growth, higher cycles and chaos

3 Numerical simulations and conjectures

3.1 Main conjecture

Our main conjecture is the following:

Conjecture 3.1.

4 Appendix

4.1 Comments on wave trains

In this section we comment on the analytical interpretation of so-called ‘wave-train’ solutions, aka stacked traveling waves or terraces.

Recall some basic facts about traveling waves:

1. A solution $u_t(x)$ is called a traveling wave with (rightward) velocity c if it satisfies

$$u_{t+1}(x) = u_t(x - c)$$

Any traveling wave has the form

$$u_t(x) = w(x - ct)$$

where w is a solution to the equation

$$Qw(x) = w(x - c)$$

We may call w a traveling wave profile with velocity c .

2. A solution $u_t(x)$ is called a periodic traveling wave with period p and mean velocity c if it satisfies

$$u_{t+p}(x) = u_t(x - cp)$$

Any periodic traveling wave has the form

$$u_t(x) = w_t(x - pc)$$

where w solves the equation

$$Q^p w_t(x) = w_t(x - pc)$$

and is periodic in t ,

$$w_{t+p} = w_t$$

We can say that w_t is a periodic traveling wave profile.

3. Alternatively we can just define a periodic traveling wave of Q to be a traveling wave of Q^p .
 4. We say that a solution $u_t(x)$ converges uniformly to a traveling wave if

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w(x - c)| = 0$$

where w_t is a traveling wave profile with velocity c .

5. We say that a solution $u_t(x)$ converges uniformly to a periodic traveling wave if

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w_t(x - pc)| = 0$$

where w_t is a periodic traveling wave profile with velocity c .

Proposition 4.1. Suppose $u_t(x)$ converges to a traveling wave, that is

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w(x - c^*t)| = 0$$

for some traveling wave profile w with velocity c^* , and suppose w has limits at $\pm\infty$ Then:

1. If $c < c^*$ then $u_t(x + ct) \rightarrow w(-\infty)$ uniformly on compact sets.
2. If $c = c^*$ then $u_t(x + ct) \rightarrow w$ uniformly.
3. If $c > c^*$ then $u_t(x + ct) \rightarrow w(\infty)$ uniformly on compact sets.

Proof. 1. Suppose $c < c^*$ and let K be a compact set. By the triangle inequality, for all x and t ,

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w(-\infty)| &\leq \sup_{x \in K} |u_t(x + ct) - w(x - (c^* - c)t)| \\ &\quad + \sup_{x \in K} |w(x - (c^* - c)t) - w(-\infty)| \end{aligned}$$

It suffices to show both terms on the right vanish as $t \rightarrow \infty$.

For the first term we have

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w(x - (c^* - c)t)| &= \sup_{x \in (K + ct)} |u_t(x) - w(x - c^*t)| \\ &\leq \sup_{x \in \mathbb{R}} |u_t(x) - w(x - c^*t)| \end{aligned}$$

which vanishes by assumption that u_t converges to a traveling wave.

For the second part, $\epsilon > 0$. We need to find t_0 such that

$$\sup_{x \in K} |w(x - (c^* - c)t) - w(-\infty)| < \epsilon, \quad \forall t \geq t_0$$

Let x_0 be such that $|w(x) - w(-\infty)| < \epsilon$ for all $x \leq x_0$. It suffices to choose t_0 such that

$$x - (c^* - c)t \leq x_0, \quad \forall x \in K, \quad \forall t \geq t_0$$

For this it suffices to have

$$\sup K - (c^* - c)t \leq x_0, \quad \forall t \geq t_0$$

Solving this inequality shows that it suffices to take any $t_0 \geq \frac{x_0 - \sup K}{c^* - c}$.

2. If $c = c^*$ then simply apply translation inside the supremum:

$$\lim_{t \rightarrow \infty} \sup_x |u_t(x + c^*t) - w(x)| = \lim_{t \rightarrow \infty} \sup_x |u_t(x) - w(x - c^*t)| = 0$$

3. The third case mirrors the first case.

□

The next theorem generalizes this to periodic traveling waves.

Remark 4.2. The next proposition assumes w_t has limits at $\pm\infty$ *uniformly in t* . This means:

$$\forall \epsilon > 0, \exists x_0, \forall t, \forall x \leq x_0, |w_t(x) - w_t(-\infty)| < \epsilon$$

(and likewise at $+\infty$).

Proposition 4.3. Suppose $u_t(x)$ converges to a periodic traveling wave, that is

$$\lim_{t \rightarrow \infty} \sup_{x \in \mathbb{R}} |u_t(x) - w_t(x - c^*t)| = 0$$

for some periodic traveling wave profile w_t with mean velocity c^* , and suppose w_t has limits at $\pm\infty$ *uniformly in t* . Then:

1. If $c < c^*$ then $u_t(x + ct) \rightarrow w_t(-\infty)$ uniformly on compact sets.
2. If $c = c^*$ then $u_t(x + ct) \rightarrow w_t$ uniformly.
3. If $c > c^*$ then $u_t(x + ct) \rightarrow w_t(+\infty)$ uniformly on compact sets.

Or if we want to be more precise:

1. If $c < c^*$ then $u_t(x + ct) - w_t(-\infty) \rightarrow 0$ uniformly on compact sets.
2. If $c = c^*$ then $u_t(x + ct) - w_t(x) \rightarrow 0$ uniformly.
3. If $c > c^*$ then $u_t(x + ct) - w_t(+\infty) \rightarrow 0$ uniformly on compact sets.

Proof. 1. Suppose $c < c^*$ and let K be a compact set. By the triangle inequality, for all x and t ,

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w_t(-\infty)| &\leq \sup_{x \in K} |u_t(x + ct) - w_t(x - (c^* - c)t)| \\ &\quad + \sup_{x \in K} |w_t(x - (c^* - c)t) - w_t(-\infty)| \end{aligned}$$

It suffices to show both terms on the right vanish as $t \rightarrow \infty$.

For the first term we have

$$\begin{aligned} \sup_{x \in K} |u_t(x + ct) - w_t(x - (c^* - c)t)| &= \sup_{x \in (K + ct)} |u_t(x) - w_t(x - c^*t)| \\ &\leq \sup_{x \in \mathbb{R}} |u_t(x) - w_t(x - c^*t)| \end{aligned}$$

which vanishes as $t \rightarrow \infty$ by assumption that u_t converges to a traveling wave.

For the second part, $\epsilon > 0$. We need to find t_0 such that

$$\sup_{x \in K} |w_t(x - (c^* - c)t) - w_t(-\infty)| < \epsilon, \quad \forall t \geq t_0$$

Let x_0 be such that $|w_t(x) - w_t(-\infty)| < \epsilon$ for all $x \leq x_0$ and for all t . It suffices to choose t_0 such that

$$x - (c^* - c)t \leq x_0, \quad \forall x \in K, \quad \forall t \geq t_0$$

For this it suffices to have

$$\sup K - (c^* - c)t \leq x_0, \quad \forall t \geq t_0$$

Solving this inequality shows that it suffices to take any $t_0 \geq \frac{x_0 - \sup K}{c^* - c}$.

2. If $c = c^*$ then simply apply translation inside the supremum:

$$\lim_{t \rightarrow \infty} \sup_x |u_t(x + c^*t) - w_t(x)| = \lim_{t \rightarrow \infty} \sup_x |u_t(x) - w_t(x - c^*t)| = 0$$

3. The third case mirrors the first case.

□

Remark 4.4. Source file for this manuscript and figures are available at github.com/mdnestor/ide-traveling-waves.

5 Introduction

This manuscript concerns periodic traveling wave solutions to the integrodifference equation

$$u_{t+1}(x) = Q[u_t](x) = \int_{-\infty}^{\infty} k(x-y)f(u_t(y))dy \quad (5.1)$$

when the growth function f has both strong Allee effect (stability at zero) and overcompensation (non-monotonicity) with a stable 2-cycle.

Traveling waves in integrodifference equations with Allee effect were analyzed in [5] and [6], the former proving existence and stability conditions and the latter giving a formula for the sign of wave speed under certain conditions. On the other hand, [4] and [1] investigated integrodifference equations with overcompensatory growth, and gave numerical evidence for a period-doubling cascade of stacked traveling waves (see Figure 2). Studies incorporating both effects include [2], [8], [3], and [7], where fluctuating wave speeds and non-spreading solutions have been observed. In particular, [8] and [7] analyzed a piecewise-constant growth function with both Allee effect and overcompensation, and the latter proved existence of a bistable 2-periodic traveling wave solution.

This manuscript focuses on the special case of the combination of Allee effect and overcompensation when the non-monotonicity induces a stable 2-cycle in the growth function. Specifically:

- (G1) $f(0) = 0$ and $f'(0) < 1$, so that $u = 0$ is a stable fixed point (strong Allee effect);
- (G2) there exists $a \in (0, 1)$ (the Allee threshold) such that $f(a) = a$ and $f'(a) > 1$, so that $u = a$ is an unstable fixed point;
- (G3) $f(1) = 1$ and $f'(1) < -1$, so that $u = 1$ is an unstable fixed point (overcompensation);
- (G4) $f(u) < u$ for all $u \in (0, a)$, $f(u) > u$ for all $u \in (a, 1)$, and $f(u) < u$ for all $u \in (1, \infty)$, so that $\{0, a, 1\}$ are the only fixed points;
- (G5) there exist $u^- \in (a, 1)$ and $u^+ \in (1, \infty)$ such that $f(u^+) = u^-$, $f(u^-) = u^+$, $f^2(u^+) = u^+$, and $f^2(u^-) = u^-$.
- (G6) $|(f^2)'(u^\pm)| < 1$, so that u^+ and u^- form a stable 2-cycle.

5.1 Traveling waves

A solution $(u_t)_{t=0}^\infty$ to equation (5.1) is called a traveling wave (or traveling front) if there exists $w \in C(\mathbb{R}, [0, \infty))$ and $c \in \mathbb{R}$ called the wave speed (or wave velocity) such that

$$u_t(x) = w(x - tc) \quad (5.2)$$

for all t and x . Additionally we assume w has limits at $\pm\infty$, which are necessarily fixed points of f . Equivalently, a traveling wave is a solution to the equation

$$Q[w](x) = w(x - c) \quad (5.3)$$

for some c . More generally, a p -periodic traveling wave is a solution to the equation

$$Q^p[w](x) = w(x - pc) \quad (5.4)$$

for some c such that w has limits at $\pm\infty$ (which are necessarily p -periodic points of f). A periodic traveling wave is called bistable if both $w(+\infty)$ and $w(-\infty)$ belong to a stable cycle of f , and monostable if only one belongs to a stable cycle.

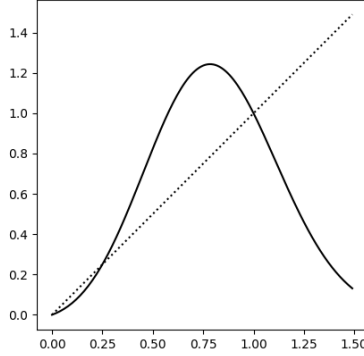


Figure 1: Example growth function satisfying our hypotheses.

5.2 Stacked waves and dynamical stabilization

A stacked wave refers to a solution of (5.1) which consists of multiple superimposed traveling waves with distinct velocities that disperse as $t \rightarrow \infty$; see Figure 2. Such solutions were observed in [4] and rigorously analyzed in [1]. They were shown to occur for the logistic growth function

$$f(u) = ru(1 - u) \quad (5.5)$$

and the Ricker growth function

$$f(u) = u \exp(r(1 - u)) \quad (5.6)$$

which both exhibit period-doubling cascades as the growth parameter r increases. A necessary (but not sufficient) condition for the existence of a stacked wave solution is a pair of traveling waves w_1 and w_2 with corresponding wave speeds c_1 and c_2 such that $w_1(+\infty) = w_2(-\infty)$ and $c_1 < c_2$.

For instance, [4] considered the logistic growth function with a stable 2-cycle paired with the Laplace kernel, and initial data of the form (6.11). The solution was shown numerically to quickly converge to a pair of stacked waves w_1 and w_2 with $w_1(-\infty) = 0$, $w_1(+\infty) = w_2(-\infty) = 1$ and $w_2(+\infty) = u^+$. This phenomenon has been referred to as "dynamical stabilization" because we have

$$u_t(x) \approx 1, \quad \forall x \in [c_1 t, c_2 t]$$

so that the solution attains the value of an unstable fixed point ($u = 1$) on an interval which grows in length as $t \rightarrow \infty$.

6 Model and simulations

We used the following 2-parameter growth function

$$f(u) = u \exp(r(1 - u)(u - a)) \quad (6.1)$$

where $r \geq 0$ and $0 < a < 1$. This map is a reparameterization of the growth map used in [3],

$$f(u) = u \exp\left(R(1 - u)\left(\frac{u}{a} - 1\right)\right) \quad (6.2)$$

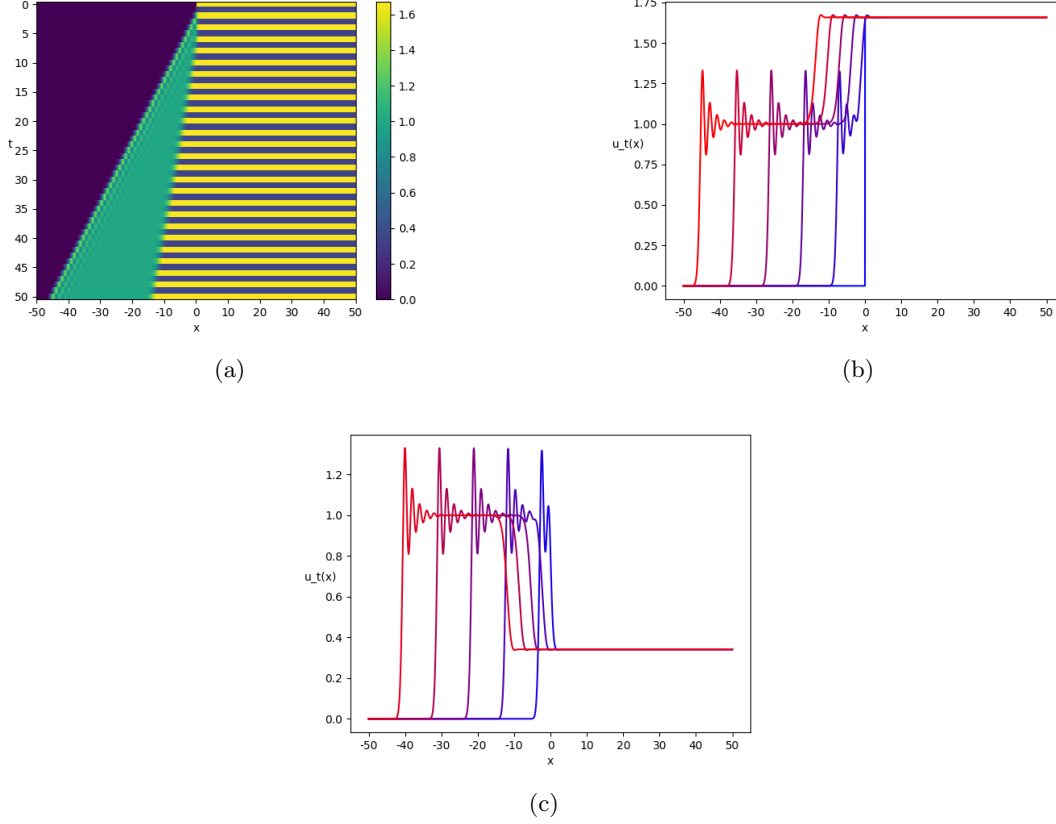


Figure 2: Convergence to stacked traveling waves using the Ricker function (5.6) with $r = 2.4$, Laplace dispersal kernel, and initial data $u_0(x) = u^+H(x)$. (a) shows the space-time heatmap of the solution. (b) shows the solution curves on time steps $t = 0, 10, 20, \dots$ while (c) shows the solution curves on $t = 5, 15, 25, \dots$, and the color changes from blue to red as t increases.

via $R = ra$.

This map has a period-doubling cascade for each choice of a ; that is, a sequence $(r_{a,n})_{n=0}^{\infty}$ (depending on a) such that, whenever $r_{a,n} < r < r_{a,(n+1)}$ then the growth map has a stable 2^n -cycle and unstable 2^k -cycles for $k = 0, \dots, (n-1)$. Our conditions are therefore satisfied whenever

$$r_{a,1} < r < r_{a,2} \quad (6.3)$$

For ease of calculation (6.9) can be expressed as

$$f(u) = u \exp(rp(u)); \quad p(u) = (1-u)(u-a) \quad (6.4)$$

Its first derivative is

$$f'(u) = (1 + rup'(u)) \exp(rp(u)) = (1 + r(a+1)u - 2ru^2) \exp(r(1-u)(u-a)) \quad (6.5)$$

We can determine its monotonicity and the stability of the fixed points by computing

1. $f'(0) = \exp(-ar) \in (0, 1)$,
2. $f'(a) = 1 + ra(1-a) \in (1, \infty)$, and
3. $f'(1) = 1 - r(1-a)$.

So, $u = 0$ is always stable and $u = a$ is always unstable. Assuming $r > 0$, we have

$$f'(1) < 0 \iff r < \frac{1}{1-a} \implies f \text{ is monotone on } [0, 1] \implies u = 1 \text{ is stable} \quad (6.6)$$

Therefore,

$$u = 1 \text{ is stable} \iff |f'(1)| < 1 \iff r < \frac{2}{1-a} \quad (6.7)$$

We have,

$$r_{a,0} = 0; \quad r_{a,1} = \frac{2}{1-a} \quad (6.8)$$

Computing $r_{a,2}$ is not straightforward and involves solving a transcendental equation $f^2(u) = u$ where $f^2 = f \circ f$ is the second-iterate map,

$$\begin{aligned} f^2(u) &= f(u) \exp(rp(f(u))) \\ &= u \exp(r(1-u)(u-a) + r(1-u \exp(r(1-u)(u-a)))(u \exp(r(1-u)(u-a)) - a)) \end{aligned} \quad (6.9)$$

Locating the 2-periodic points thus amounts to solving the equation

$$f^2(u) = u \iff p(u) + p(u \exp(rp(u))) = 0 \quad (6.10)$$

In practice, assuming $r_{a,1} < r < r_{a,2}$, a numerical scheme can be used to estimate $u^{\pm}$ by perturbing the unstable fixed point.

The initial data is given by the step function

$$u_0(x) = \begin{cases} 0 & x < 0 \\ u^+ & x \geq 0 \end{cases} \quad (6.11)$$

The model with initial condition (6.11) exhibits two distinct behaviors, depending on the model parameters, as illustrated below in Figure 3. Either:

Outcome 1. a single bistable 2-periodic traveling wave connecting $u = 0$ to $u = u^\pm$ (as shown in the left column of Figure 3);

Outcome 2. a pair of monostable traveling waves that form a stacked wave train, connecting $u = 0$ to $u = 1$ and $u = 1$ to $u = u^\pm$, respectively (as shown in the right column of Figure 3).

This observation naturally leads to several questions:

- Q1. for what regions of the parameter space does dynamical stability/a stacked traveling front occur, vs a periodic traveling wave?
- Q2. can we prove existence/stability of the wave in either case?
- Q3. is there a general formula, as in [6], that could guarantee/forbid the existence of this wave? For example, if there is a wave w_1 with $w_1(-\infty) = 0$ and $w_1(+\infty) = 1$ and a periodic traveling wave w_2 with $w_2(-\infty) = 1$ and $w_2(+\infty) = u^+$, with speeds c_1 and c_2 respectively, is $c_1 > c_2$ a necessary/sufficient condition?

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A example

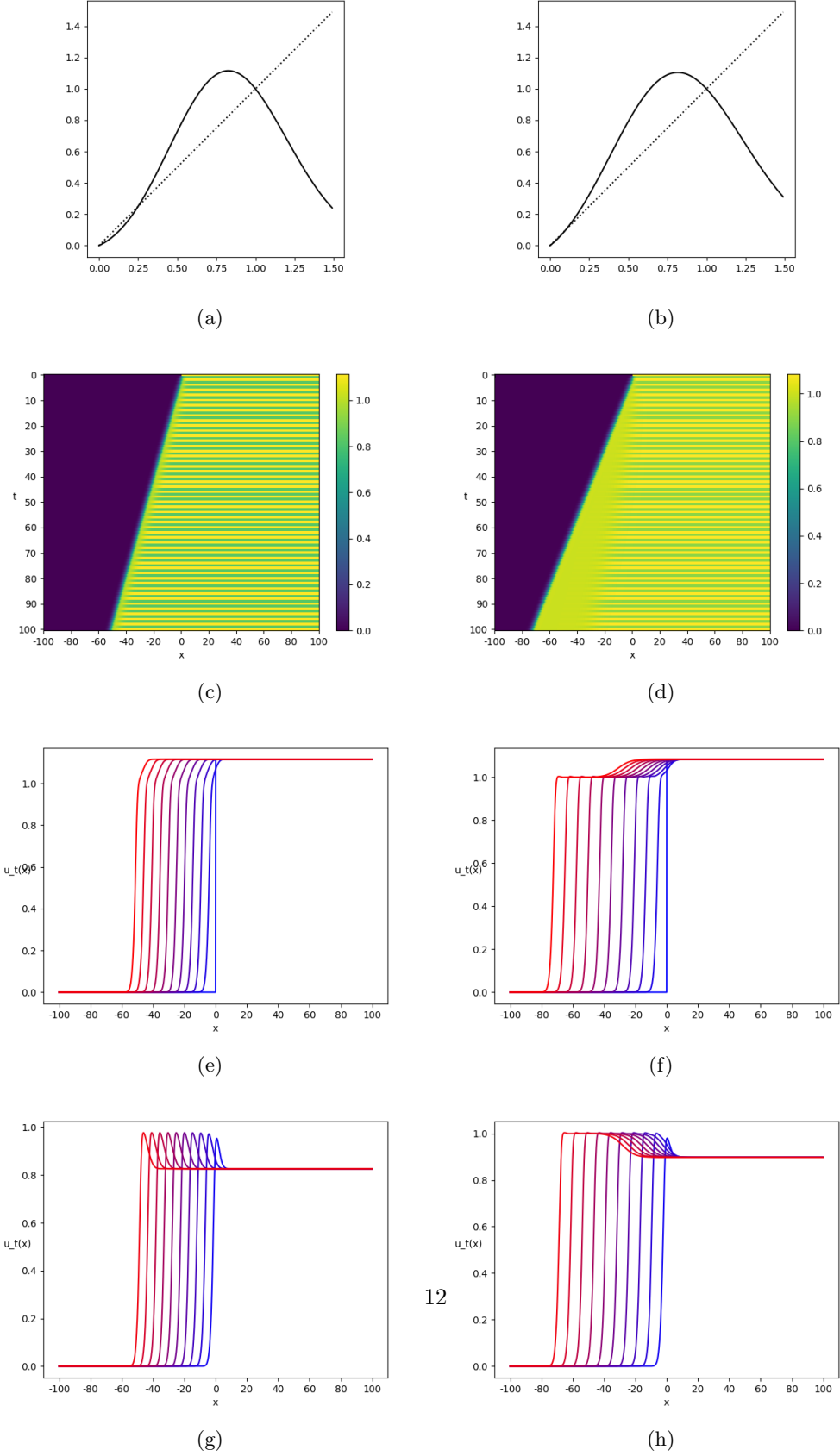


Figure 3: The left and right columns show our growth function (6.9) with parameters ($a = 0.25, r = 3.0$) and ($a = 0.1, r = 2.3$) respectively. (a) and (b) show the plot of the growth curve, while (c) and (d) show convergence to a pair of stacked waves vs. a single 2-periodic traveling wave, respectively.